

GOURAB ROY

Pázmány Péter Catholic University, Budapest, Hungary

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RESEARCH AREA

Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Medical Image Analysis, NLP

EDUCATION

•Pázmány Péter Catholic University, Budapest, Hungary

Erasmus Mundus Joint Master's Degree in Artificial Intelligence for IPCV

September 2025 – Present

•Sylhet Engineering College, Sylhet, Bangladesh

B.Sc.(Engg.) in Computer Science and Engineering

February, 2018 - January, 2023

CGPA: **3.56**/4.00

PUBLICATIONS

•Live

- Ornob, T.R., **Roy, G.** & Hassan, E. (2022). “CovidExpert: A Triplet Siamese Neural Network framework for the detection of COVID-19”. Informatics in Medicine Unlocked, 101156, (Jan 13, 2023). DOI: 10.1016/j.imu.2022.101156 ([Link](#))
- Islam, T., Omur, S.K.H., Nahid, N., Chowdhury, L., **Roy, G.**, Mojumder, M.A.N., Soeb, M.J.A. and Jubayer, M.F., 2025. **Optical character recognition for ancient scripts: a case study on Syloti Nagri using deep learning models.** Journal of Computer Sciences Institute, 34, pp.98-107. ([Link](#))
- Islam, K.S., **Roy, G.**, Nahid, N., Ripti, S.Y., Mojumder, M.A.N., Soeb, M.J.A. and Jubayer, M.F., 2025. **Advancing Bangla typography: machine learning and transfer learning based font detection and classification approach using the ‘Bang-laFont45’ dataset.** Journal of Computer Sciences Institute, 35, pp.166-174. ([Link](#))
- Gazi, M.U., Rupom, N.I., Durjoy, G.B., **Roy, G.**, Nahid, N., Hossain, M.M. and Hasan, M.T., 2025, February. **Enhancing Concrete Performance: Machine Learning Models for Predicting and Classifying Tensile Strength.** In 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) (pp. 1-6). IEEE. ([Link](#))
- Gazi, M.U., Durjoy, G.B., **Roy, G.**, Nahid, N., Hossain, M.M. and Saha, G., 2024, December. **CrackClassification4: Classifying Four Types of Cracks in Structural Surfaces Using CNNs.** In 2024 27th International Conference on Computer and Information Technology (ICCIT) (pp. 2506-2511). IEEE. ([Link](#))
- Hossain, M.M., Nahid, N., **Roy, G.**, Mojumder, M.A.N. and Rahman, M.S., 2024, December. **Optimizing Multilingual Summarization: Advanced Corpus Creation and Filtering Approach.** In 2024 27th International Conference on Computer and Information Technology (ICCIT) (pp. 2482-2487). IEEE. ([Link](#))

•Undergrad Thesis

- CerebralNet: A deep learning model to detect major abnormalities from brain CT and MRI scan via segmentation and classification techniques.

•Accepted & Under Review

- Intelligent Infrastructure: Predicting Bridge Collapse with Advanced Machine Learning (*Accepted*)
- Deep Learning-Based Detection of Aircraft Cracks and Dents Using YOLOv11 and YOLOv12 for Aviation Safety in Bangladesh (*Accepted*)
- BanglaRiceGrain-5: A Comprehensive Dataset for Classifying Bangladeshi Rice Varieties Using Deep CNN Architectures (*Under Review*)
- Tomato lead diseases detection using YOLO Architectures (*Working on*)

EXPERIENCE

•Research Assistant

November 2023 – Present

Sylhet Engineering College

Sylhet, Bangladesh

- Work under the supervision of the Head of the Computer Science and Engineering Department, **Md. Abu Naser Mojumder**.
- Engaged in the department's Research Lab to create machine learning solutions for various applications.
- Provide guidance to current students and supervise final year academic thesis projects.

•Adjunct Lecturer

January 2024 – Present

Sylhet Engineering College

Sylhet, Bangladesh

- Responsible for taking undergraduate classes, marking assessments, consulting students, conducting research, and supervising final year thesis students.
- Courses instructed:
 - * Introduction to Distributed Computing
 - * Introduction to Distributed Computing (Sessional)
 - * Computer Architecture
 - * Introduction to Data Science
 - * Introduction to Data Science (Sessional)
 - * Peripheral and Interfacing
 - * Microprocessor & MicroController
 - * Microprocessor & MicroController (Sessional)

PROJECTS

•Data Analysis Projects

[Source Code](#)

Python, Numpy, Matplotlib, Pandas, Jupyter Notebook

- Developed expertise in exploratory data analysis and visualization by completing diverse projects, including analysis of **Boston housing data**, **Covid-19 global health outcomes**, and **Google Play app ratings data**.
- Created a Git repository featuring **8 exploratory data analysis projects**, demonstrating skills in data visualization and analysis.

•Python Projects

[Source Code](#)

Python, PyCharm, Jupyter Notebook

- Developed multiple Python projects, including **Calculator**, **Dice rolling simulator**, **Tic Tac Toe game**, **Caterpillar game**, **Egg Catcher game**, and **Hangman game**.

•Java Application

University Project

Java, NetBeans IDE, MySQL, JFrame

[Source Code](#)

- Designed and developed a **Billing System application** using **Java and MySQL** for university coursework, showcasing skills in programming, database design, and software development.

•Web-Based Project

[Source Code](#)

HTML, CSS, JavaScript

- Designed and developed a simple website called **Pioneer Bank** using HTML, CSS, and JavaScript.
- Published the project on **GitHub** for public access.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript

Machine Learning Libraries: PyTorch, TensorFlow, Scikit-learn

Databases: MySQL, SQLite

Developer Tools: Visual Studio Code, PyCharm, NetBeans IDE

Other Skills: Git, Windows, Linux, LaTeX

IELTS SCORE

Overall Band: 6.5, Listening: 7, Reading: 6.5, Writing: 6, Speaking: 6.0

CERTIFICATIONS & ACHIEVEMENTS

Problem Solving: Solved 350+ problems on platforms such as [HackerRank](#), [Codeforces](#), [UVA](#)

Achievements: Earned Gold Badges in C++ and Python problem solving on HackerRank

Machine Learning Specialization: Completed Machine Learning Specialization course on Coursera ([Link](#))

Data Structures Course: Completed Data Structures course on Coursera ([Link](#))

Data Visualization Masterclass: Completed Data Visualization in Python Masterclass on Udemy ([Link](#))

Review Contribution: Contributed as a reviewer for Informatics in Medicine Unlocked ([Link](#))